

Ritesh Mamidi

Applied AI Systems · Agent Evaluation · Data & Product Analytics

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SUMMARY

Data and AI/ML-quality professional with 4+ years in data and analytics, currently validating LLM/ML outputs and investigating failure modes. Independently design and evaluate open-source applied-AI systems spanning agentic workflows, tool-calling agent infrastructure, retrieval and human-in-the-loop controls using Python, SQL, PostgreSQL and FastAPI.

PROFESSIONAL EXPERIENCE

Data & AI Analyst — Apple (client engagement) · Austin, TX Sep 2025 – Present

- Validate LLM/ML outputs against policy and quality rubrics, investigating edge cases and systematic failure patterns.
- Build QA/QC frameworks and operational reporting with Python, SQL and Power BI, contributing to an 18% improvement in investigation efficiency and a 30% reduction in operational escalations.
- Apply LLM-assisted analysis and prompt engineering to classify cases and extract operational signals.

Business / Data Analyst — Heart Hospitality LLC · United States Jan 2024 – Aug 2025

- Built Python/SQL ETL pipelines and automated reporting, reducing manual reporting effort 55%.
- Designed relational data models and forecasting/classification models that improved planning accuracy 22%.

Data / Business Analyst — Liberated Monks Pvt. Ltd. · India Jan 2022 – Dec 2023

- Built Python/SQL ETL pipelines and Tableau dashboards; owned data validation and KPI definitions.
- Built ML segmentation and propensity models associated with an 18% improvement in marketing-driven engagement.

INDEPENDENT AI ENGINEERING

Independent open-source work · synthetic data/tasks · not employer/client work

Support Escalation Copilot — Agentic AI application · Python · PostgreSQL · pgvector

- Designed an approval-gated agentic workflow combining retrieval, structured LLM outputs, typed actions and deterministic policy, with PostgreSQL row-level security for tenant isolation.
- Designed and verified human approvals bound to exact actions, idempotent execution, crash recovery and tamper-evident auditing; tested control boundaries through adversarial and mutation testing.
- Evaluated a frozen protocol with a local Qwen3-4B model and model-free replay, identifying structured-output/interface failures while separating model performance from deterministic control behavior.

AI Agent Runtime + Agent Runtime Benchmark — Agent infrastructure · Python · FastAPI · PostgreSQL · Docker

- Designed a Python agent runtime with a model-tool loop, schema-validated tools, permissions and approvals, bounded retries/budgets, tracing, memory and planning modules.
- Designed and verified a FastAPI/PostgreSQL service layer with persistent run state, approvals, resumable event streaming and restart-durability testing in Docker Compose CI.
- Evaluated a local 4B model through native tool calling on a frozen 48-task benchmark, separating model behavior from runtime controls; surfaced an information-flow failure through a permitted read tool, showing that action safety does not equal information-flow safety.

LLM & Agent Evaluation — MAREF + Ilmeval · Python

- Designed evaluation tooling for LLM outputs (Ilmeval: deterministic metrics, weighted rubrics, hard gates) and agent runs (MAREF: eight dimensions, fixed failure taxonomy, frozen protocol).
- Pre-registered and evaluated the MAREF research prototype on a held-out split; reported a MIXED result rather than collapsing failure detection and false alarms into a single favorable score.

SKILLS

AI systems: LLM/GenAI integration (local models), AI agents and agentic workflows, tool/function calling, structured outputs, prompt engineering, RAG/vector retrieval with embeddings, guardrails, human-in-the-loop systems

Evaluation: LLM/agent evaluation, benchmark design, failure analysis, frozen and pre-registered protocols, reproducible replay

Engineering: Python, SQL, PostgreSQL, FastAPI, REST APIs, Docker, CI/CD (GitHub Actions), pytest, Git

Analytics: Power BI, Tableau, ETL, KPI design, machine learning, scikit-learn, pandas, experimentation & A/B testing

EDUCATION & CERTIFICATIONS

MS, Business Analytics — St. Francis College (US)

M.Tech (Integrated), Software Engineering — Vellore Institute of Technology (VIT), India

IBM AI Engineering Professional Certificate · Google Advanced Data Analytics Professional Certificate (Coursera)